

Details of vSphere HA and DRS-26

VMware vSphere provides enterprise-level features that ensure high availability and efficient use of resources in a virtualized environment. Two of the most important features are **High Availability (HA)** and **Distributed Resource Scheduler (DRS)**.

vSphere High Availability (HA)

HA is designed to keep virtual machines (VMs) running in case of a host failure. •

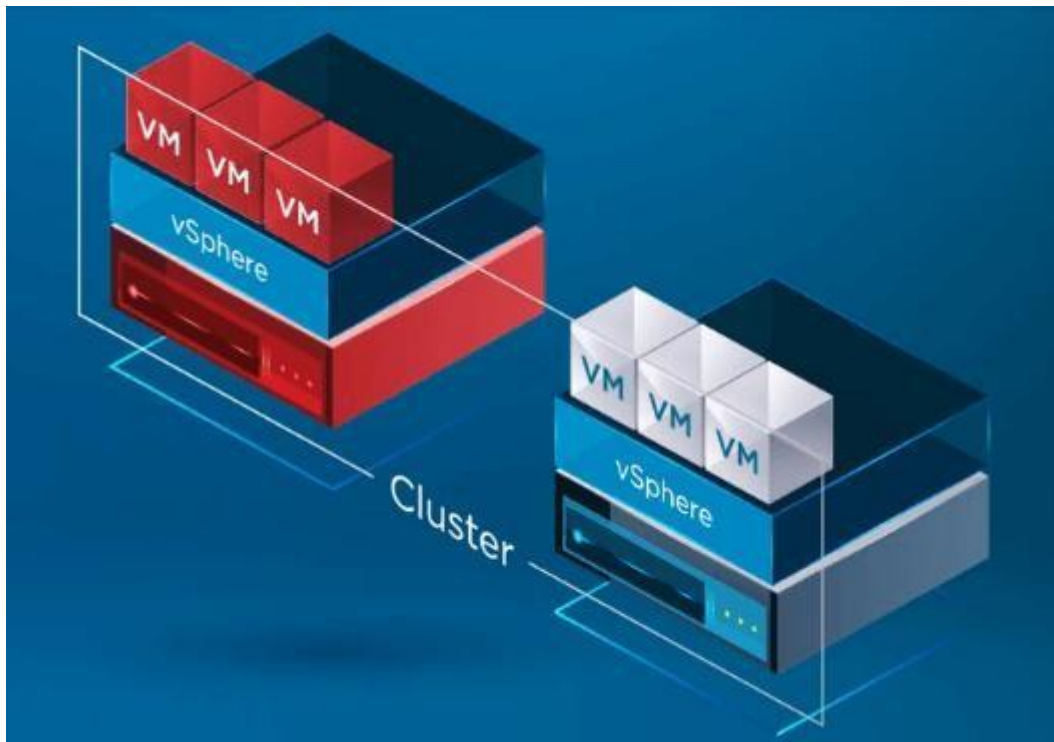
Purpose: Reduce downtime when a physical host fails.

- HA is a feature that provides automatic restart of virtual machines (VMs) on other hosts in case of host failure, while DRS is a feature that balances VMs across multiple hosts in a cluster based on resource utilization
- HA ensures VM availability during host failures, while DRS helps with resource optimization and load balancing in a cluster
- HA and DRS can work together to provide enhanced availability and resource management for VMs in a vSphere environment
- **How it works:**
 - Hosts in a cluster share a **heartbeat** through the vCenter. ◦ If a host stops sending heartbeats, HA assumes it failed.
 - HA restarts the affected VMs on other healthy hosts in the cluster.
- **Requirements:**
 - vCenter Server.
 - Shared datastore (like NFS, iSCSI, or SAN). ◦ At least two ESXi hosts in a cluster.
 - At least 4 GB of memory configured on each host ◦ vSphere Standard License
 - Pingable gateway or another reliable network node
- **Benefits:**
 - Automatic failover without admin intervention.
 - No need to manually power on VMs after host crash. ◦ Reduces downtime for critical applications.

Overview of vSphere HA Behavior

By using the benefits provided to the ESXi hosts in a vSphere Cluster, in its most basic form, vSphere HA implements a monitoring mechanism between the hosts in the vSphere Cluster. The monitoring mechanism provides a way to determine if any host in the vSphere Cluster has failed.

In the infographic below, a two-node vSphere Cluster has experienced a failure of one of the ESXi hosts in the vSphere Cluster. The vSphere Cluster has vSphere HA enabled at the cluster level.



Example:

If Host-1 fails and had 5 running VMs, HA will automatically restart those 5 VMs on Host2 or Host-3.

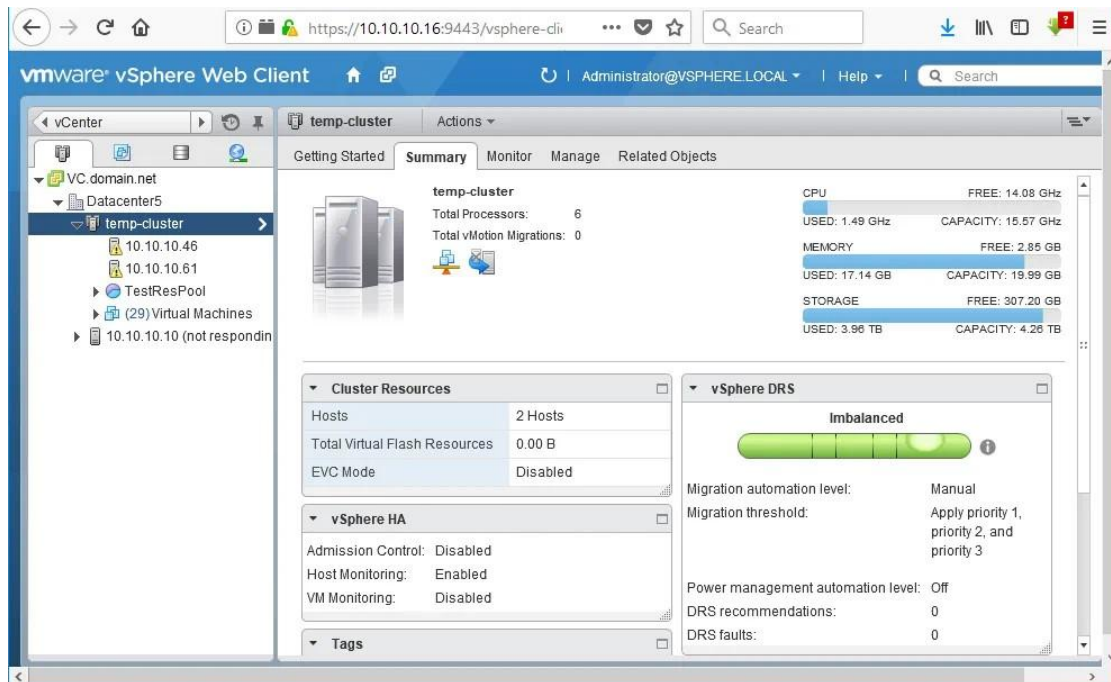
vSphere Distributed Resource Scheduler (DRS)

DRS ensures balanced performance across all hosts in a cluster.

- **Purpose:** Distribute workloads to avoid resource bottlenecks.
- **How it works:**
 - Monitors CPU and memory usage across the cluster.
 - Uses **vMotion** to move VMs between hosts when load imbalance is detected.
 - Operates in **three modes**:
 - **Manual:** Admin decides whether to apply DRS recommendations.
 - **Partially Automated:** DRS places VM at power-on but asks for approval before migrating running VMs.
 - **Fully Automated:** DRS automatically balances VMs without asking.
- **Requirements:**

- vCenter Server. ○ Shared storage for vMotion.
- Proper networking setup.
- **Benefits:**
 - Prevents resource contention. ○ Improves VM performance.
 - Reduces manual workload for administrators.

Here you can check the state of your newly created DRS cluster. Just go to vCenter -> Clusters -> Your cluster name and click the “Summary” tab.



Example:

If Host-2 is overloaded and Host-3 is underutilized, DRS automatically migrates some VMs from Host-2 to Host-3 to balance the load.

VMware vSphere HA Cluster Failure Types

In a vSphere HA enabled cluster, there are three types of failures that can happen to trigger a vSphere HA failover event. Those host failure types are:

- **Failure** – A failure is intuitively what you think. A host has stopped working in some form or fashion due to hardware or other issues.
- **Isolation** – The isolation of a host generally happens due to a network event that isolates a particular host from the other hosts in the vSphere HA cluster.
- **Partition** – A partition event is characterized by a subordinate host losing network connectivity to the master host of the vSphere HA cluster.

